on the density of the liquid. This property of a moving fluid to oppose the relative motion between its layers is called viscosity.

Cause of viscosity:

Consider a liquid flowing over a horizontal surface with two neighboring layers. The upper layer tends to accelerate the lower layer and in turn, the lower layer tends to retard the upper layer. As a result, a backward tangential force is set-up. This tends to destroy the relative motion. This accounts for the viscous behaviour of fluids.

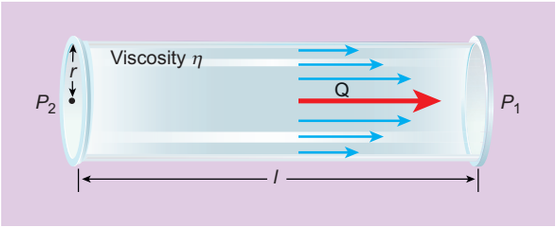


Figure 7.14 Viscosity

Coefficient of viscosity:

Consider a liquid flowing steadily over a horizontal fixed layer (Figure 7.15). The velocities of the layers increase uniformly as we move away from the fixed layer. Consider any two parallel layers A and B. Let v and $v + dv$ be the velocities of the neighboring layers at distances x and $x + dx$ respectively from the fixed layer.

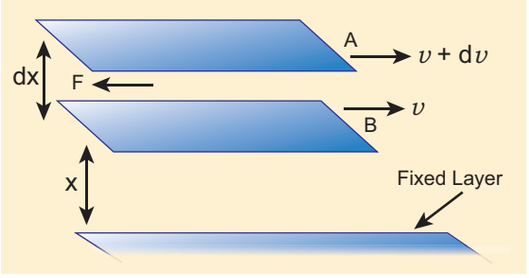


Figure 7.15 Flow of liquid over the horizontal layers

The force of viscosity F acting tangentially between two layers is given by Newton's First law. This force is proportional to (i) area A of the liquid and (ii) the velocity gradient $\frac{dv}{dx}$

$$F \propto A \text{ and } F \propto \frac{dv}{dx}$$

$$\Rightarrow F = -\eta A \frac{dv}{dx} \qquad (7.20)$$

Where the constant of proportionality η is called the coefficient of viscosity of the liquid and the negative sign implies that the force is frictional and it opposes the relative motion. The dimensional formula for coefficient of viscosity is $\left[ML^{-1}T^{-1}\right]$.

Note

Viscosity is similar to friction. The kinetic energy of the substance is dissipated as heat energy.

EXAMPLE 7.9

A metal plate of area 2.5×10^{-4} m^2 is placed on a 0.25×10^{-3} m thick layer of castor oil. If a force of 2.5 N is needed to move the plate with a velocity 3×10^{-2} m s^{-1}, calculate the coefficient of viscosity of castor oil.

Given: A=2.5×10^{-4} m^2, dx = 0.25×10^{-3}m, F=2.5 N and $dv = 3\times10^{-2}$ m s^{-1}

Solution

$$F = -\eta A \frac{dv}{dx}$$

In magnitude, $\eta = \frac{F}{A}\frac{dx}{dv}$

$$= \frac{(2.5N)}{(2.5\times10^{-4}\text{m}^2)}\frac{(0.25\times10^{-3}\text{m})}{(3\times10^{-2}\text{m s}^{-1})}$$

$= 0.083\times10^{3}$ N m^{-2}s

UNIT-7(XI-Physics_Vol-2).indd 66 02-01-2023 14:39:56